Teach for understanding and independent transfer

Programme home · Teacher guides

Purpose and limits

This is an evidence-informed teaching design applied to all 252 main weeks and 12 Python readiness lessons. It preserves the full lesson, teacher and workbook model. No classroom trial has established that it is universally best; technical checks do not demonstrate learning impact.

Teach one connected cycle

Start with three 45-minute sessions: understand, practise, investigate and transfer. Each weekly teacher guide gives the staged allocation. Schedule more sessions for connected topics, laboratories and project construction. Use learner responses to adjust pace; the starting allocation is not a validated duration.

Retrieve and model

Ask for a closed-note explanation before instruction, without grading a rough initial answer as ability. Weekly prompts revisit ideas approximately one, three and six teaching weeks earlier. Model a full worked case aloud: identify supplied facts, choose a procedure, trace steps and check the conclusion. Match words to the visual rather than treating pictures as decoration.

Fade support and check everyone

Use the named W2 completion task. Supply inputs and the deciding rule, then let learners complete the middle steps. If necessary, demonstrate one step and withdraw help on the next. Each learner answers the case-based misconception question before partner discussion. Require an explanation of the deciding fact, not only a vote. Re-model widespread errors; provide targeted support for smaller groups.

Make transfer independent

The selected Independent case in each week has its own teacher answer. First curriculum occurrences

use the primary source W4; repeated sources have new parallel cases; all 12 Python bridge lessons have changed readiness cases. Older transfer questions remain practice. Check prior exposure outside this programme and withhold feedback until the attempt is recorded. Different wording alone does not ensure a fresh case. Repeated copies in lesson/workbook share one attempt.

Assess explanation, trace and limits

For each dimension record not yet, with support or independent. Accept a defensible alternative method. Preserve the given inputs, distinguish predicted from executed output, and require evidence for claims. A supported correct answer is practice evidence; collect an independent changed case later. An honestly reported system failure can accompany excellent reasoning.

Give feedback learners act on

Identify the first unsupported step, prescribe one repair, and have the learner perform and explain it. Check the relevant skill on a further fresh case after solving that case yourself. Do not substitute praise, confidence ratings, page completion or attractive presentation for reasoning evidence.

Offer access without lowering the goal

Read aloud, enlarge visuals, allow oral explanations or drawings, and reduce redundant examples when these remove barriers. Keep the reasoning goal. Do not assign fixed visual/auditory learning styles. Extend secure learners through changed assumptions or boundaries, with a justified prediction.

Return after a delay

Weekly plans name suggested future reviews at 1, 3, 6 teaching-week intervals where the level permits. These intervals are scheduling choices, not a claimed optimal research prescription. Adjust for holidays and actual dates. At level endings, plan a return beyond the level. Retrieve the idea and apply it to a changed case; record help and the next action.

Use the complete project model

Retain every named topic and original section. Collect source records, input/output traces, controlled tests, errors, revisions and limits in the project portfolio. Require individual explanations alongside group construction. Use the 12 Python lessons unless equivalent readiness is independently demonstrated.

Example: Foundations Week1

Model the bus case: a learned delay estimate is separate from the written rule that displays red above 10, and a driver override is human action. Supported practice uses the named photo-classifier task. The independent case uses a greenhouse: distinguish its learned forecast, written CHECK PLANTS rule above 30, and caretaker action. A correct message colour does not prove an accurate forecast. Check the actual weekly teacher answer before teaching.

Pilot before making impact claims

Use the blank pilot record for real learners. Record prior exposure, baseline reasoning, support and independent outcomes. Return after a delay with a matched changed case. Compare explanation, trace and limits, and note time and accessibility barriers. Adjust weak explanations using these responses. A before/after classroom comparison alone does not establish causation; do not invent learner results.

Teach with the reviewed pictures

Every week now adds a concrete worked case, a labelled pictorial trace, a changed-condition comparison, a specific explanation question and a reasoned teacher key. Additional plots, grids and weighted models use the actual case values. Match each visual step to the supplied facts. These completed examples support modelling and practice; keep the separately selected independent case for a first attempt before feedback.

The academy end goal

Learners should understand, design, build, test and explain AI systems responsibly. Each level ends in a cumulative project with individual explanation, a transparent starting method, appropriately separated evaluation, documented failures and justified limits. Judge progress through those demonstrated skills rather than page completion or presentation alone.

Research informing the design

The IES practice guide recommends spaced learning, worked examples with practice, graphics with verbal explanations, retrieval and explanatory questions; evidence strength differs by recommendation. EEF guidance supports subject-specific planning, monitoring and evaluation, explicit modelling and scaffolding towards independence, and feedback that leads to action. These sources support design principles, not a claim that this complete programme has proven effectiveness.

- IES: Organizing Instruction and Study to Improve Student Learning
- EEF: Metacognition and Self-Regulated Learning
- EEF: Teacher Feedback to Improve Pupil Learning

Blank classroom pilot record

Practice and keep evidence

Use supported questions, hints and changed cases to practise a worked explanation. Record completed answers and reflections, then download them using the portfolio controls. Practice feedback does not replace independent assessment.

Supported Python examples run in a browser worker with real output and a five-second execution limit. Predict first, run, inspect errors, change one thing and test again. Earlier lesson links provide suggested repair tasks.

Offline Python needs a local HTTP server: run `python3 -m http.server 8000 --bind 127.0.0.1` from the programme folder, open the local student index, and stop the server after use. Reading and paper tracing remain available when opening files directly.

Run a real classroom pilot

Use the teacher pilot studio on the website or the offline pilot guide. Three readiness tracks offer baseline, independent post-lesson and delayed tasks. Record real anonymous learner evidence, support, exposure, time and barriers before revising the lesson. No real classroom outcomes have yet been collected.